# Justin Xu

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## EDUCATION

**Georgia Institute of Technology** — Atlanta, GA
*Bachelor of Science in Computer Science* — Expected Graduation Date: May 2027
Concentrations: Intelligence and Modeling and Simulation — GPA: 3.68

## SKILLS

- **Programming Languages:** Python, Java, C++, C, Typescript, Javascript, SQL, Bash, R, HTML, CSS
- **Operating Systems/Platforms:** Windows, Linux, MacOS, Azure, Kubernetes, Openshift, Docker, Raspberry Pi
- **Frameworks:** PyTorch, LangChain, RAG, Flask, React Native
- **Software/Packages/Tools:** NumPy, SciPy, Matplotlib, Pandas, Keras, Scikit-Learn, ChromaDB, MySQL, FAISS, Transformers, Jupyter Notebook, Git/Github, LAMMPS, Android Studio, Junit, Firebase, Ansible, VS Code, Cursor, Claude, Power BI

## WORK EXPERIENCE

**IBM** — Brookhaven, GA
*Customer Success Engineer Co-op (Data & AI)* — August 2026 - Present

- Using Python, open source frameworks, RedHat, and IBM watsonx to architect and develop proof of concept and MVP (minimum viable products) builds to drive enterprise adoption and deployment of IBM's data and AI solutions.

**Novelis** — Kennesaw, GA
*Research and Development Engineer Intern* — June 2023 – July 2023

- Worked with a team to develop a robotic autonomous system for transporting and administering a three-point bend test on aluminum alloy samples on a MTS test fixture using a UR16e robotic arm.
- Designed a system in CAD with custom fabricated CNC plates, 2 step motors, limit switches, and a custom PCB.
- Programmed the system's hardware control logic in Python using a Raspberry Pi to autonomously adjust the width of the MTS test fixture based on sample size using 20+ GPIO pins.

## RESEARCH EXPERIENCE

**Georgia Tech Vertically Integrated Projects** — Atlanta, GA
*Undergraduate Researcher in the Surrogate Modeling for Urban Regeneration (SMUR) team* — January 2025 - May 2026

- Developed agent-based simulation software in Python that uses game theory and no-regret dynamics to simulate the demographic changes of the Atlanta Beltline.
- Processed 13 years of Census and OpenStreetMap (OSM) data via APIs and data pipelines to train 1000+ simulated agents moving across 250+ census tracts across Metro Atlanta.
- Developed automated test scripts for error calculation, convergence detection, and hyperparameter optimization and calibration across 100+ simulation runs.

**Georgia Tech Bhamla Lab** — Atlanta, GA
*Undergraduate Researcher* — June 2024 – May 2025

- Worked on a research project on active polymers using LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator) Brownian physics simulation software.
- Used Python data analysis packages such as NumPy, SciPy, and Matplotlib, and bash scripts to run 1500+ simulation iterations on Georgia Tech's HPC research clusters, analyzing and processing 100,000+ data points.
- Co-authored and published a paper in the journal Physical Review Letters (PRL): https://doi.org/10.1103/ntm6-dz86

## PROJECTS

**Glasses Notes**

- An automated AI-powered presentation note-taking app for Ray-Ban Meta smart glasses, built with Typescript, Javascript, React Native, and deployed using Expo.
- Includes features for automated audio/slide transcription and organized summarization/note-taking using Whisper and GPT.